Positional Mechanisms in Attention:
two slots, one scan, and what each one is for

Abstract

Attention encodes no order by itself, and the mechanisms proposed to supply it now number in the dozens. This review argues the space is far smaller than the list.

If the positional map is linear in the query and key, translation-invariant and continuous, the interval operators form a one-parameter matrix group $A(d) = \exp(dM)$, so classifying positional encodings reduces to classifying the generator M up to conjugacy — which leaves exactly two live slots, a real part that scales and an imaginary part that rotates, because $\mathbb{C}$ has exactly two. Defective generators add polynomial factors and nothing else. Writing both slots as one complex log-polar accumulator makes every mechanism a choice of where content enters: magnitude or phase, instantaneous or accumulated, at what sign and what rank. Mechanisms that appear to escape — PaTH, DeltaNet, RWKV-7, CoPE, DAPE, MesaNet, Titans, TAPE, TEM-t, and the classical relative methods of Shaw, Transformer-XL and T5 — escape by breaking one of the three assumptions, and it is possible to say which.

Two consequences do not follow from any single mechanism. The two slots answer *different questions*: a signed accumulated phase measures net displacement and is a map, a monotone one measures elapsed path and is a clock, and the two are mutually exclusive — so the monotone increments of CARoPE, GRAPE-AP and CoPE are correct on language and disqualifying on navigation. That the distinction is invisible on text is not incidental: in one content-dependent mechanism's own ablation, NoPE attains the lowest perplexity of every encoding tested. And the frame has a second reading, in which the positional operator is an associative memory's transport kernel; under it the fast-weight literature and the positional-encoding literature are seen to have divided the two slots between them without either saying so.

The review closes with measurements on navigation and algorithmic tasks isolating the two axes the frame exposes and nobody varies deliberately — the *sign* of the increment, which costs nothing and is worth more than any other ingredient here, and the *input-side rank* of the content-to-phase map — and with what the two imply for choosing a mechanism for a given task.

1 Introduction

Attention computes content-dependent interactions between tokens and, by itself, encodes no order at all: permute the inputs and the outputs permute with them. A positional mechanism is whatever is added to restore order, and the space of such mechanisms is now large — absolute embeddings, relative biases, rotary phases, decays, gates, content-dependent variants of each, and their state-space analogues.

This review argues that the space is much smaller than the list suggests.

The thesis. If the positional map is linear in the query and key, translation-invariant, and continuous, then the family of interval operators is a one-parameter matrix group $A(d) = \exp(dM)$, and classifying positional encodings reduces to classifying the generator M up to conjugacy. Diagonalising M leaves exactly two live slots — a real part that scales and an imaginary part that rotates — because $\mathbb{C}$ has exactly two. Writing both as one complex log-polar accumulator turns every mechanism in the arena into a choice of *where content enters*: in the magnitude or the phase, instantaneously or accumulated, with what sign and at what rank. Mechanisms that appear to escape this picture escape by breaking one of the three assumptions, and it is usually possible to say which.

What the frame buys. Two things a list of mechanisms does not. First, the two slots turn out to answer *different questions* rather than to be two settings of one dial: a signed accumulated phase measures net displacement and is a map, while a monotone one measures elapsed path and is a clock, and the two are mutually exclusive by construction (§8). Second, the frame makes visible two axes on which the mechanisms genuinely differ and which nobody varies deliberately — the *sign* of the increment and the *input-side rank* of the content-to-phase map — and both are invisible on the tasks the literature evaluates.

Scope, and what this review is not. It covers positional mechanisms as operators on the attention logit: how position enters, what algebra it lives in, and what that implies for what can be computed. It does not cover the frequency-allocation and context-extension line (Position Interpolation, NTK-aware scaling, YaRN, LongRoPE), which varies the readout rather than the operator and is surveyed elsewhere; §12 states only what that line contributes to the frame. Original measurements are reported in §13 and are marked as such throughout.

Positioning. Four surveys of this material were checked. Three — Li (arXiv:2608.10021, August 2026) on absolute and relative methods, RoPE and long-context scaling; the length-extrapolation survey (arXiv:2312.17044); and the infinite-extrapolation framework of Vetcha (arXiv:2601.06113) — contain no content-dependent *phase* mechanism at all: between them, zero occurrences of MapFormer, Selective RoPE, CARoPE, FoX, GRAPE or Mamba-3, and CoPE and Mamba twice each. The fourth, Zhu et al.'s long-context survey (arXiv:2503.17407, March 2025), *does* carry a category for it, "content-aware position embedding", with two members: CoPE and DAPE. That category is the closest existing treatment of this material, and it predates almost all of it — CARoPE by four months, Selective RoPE, MapFormer and GRAPE by eight to nine, Mamba-3 by a year.

So the gap is not that nobody noticed content-dependence. It is that the cell was opened on the *additive* side, by DAPE (2024) and CoPE (2024), and everything that has since filled the *multiplicative* side arrived after the last survey that had a place to put it. The memory reading of §11 and the navigation regime appear in none of the five.

Organisation. §2–3 set up the polar frame and the classification. §4 places the classical relative methods, which fail the classification's third condition. §5–6 treat the two slots in turn and §7 unifies them in one equation. §8 is the conceptual consequence. §9 is the relational map over eight axes; §10 places two mechanisms from the cognitive-map literature and §11 re-reads the whole frame from the fast-weight side. §12 covers what the long-context line contributes, §13 reports the measurements, and §14 draws out what they imply for choosing a mechanism.

2 The shared frame

Rotary attention transforms the query and key of head h by a block-diagonal rotation before the inner product. Splitting a head's d_{head} coordinates into $n_b = d_{\text{head}}/2$ planes and identifying each plane with $\mathbb{C}$, write $\tilde{q}_t^{(c)}, \tilde{k}_s^{(c)} \in \mathbb{C}$ for the query at position t and key at position s in plane c. The pre-softmax logit is

$$a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=1}^{n_b} \mu_{t,c}^{q}\, \mu_{s,c}^{k}\, \cos\bigl(\theta_s^{(c)} - \theta_t^{(c)} + \psi_{s,c}^{k} - \psi_{t,c}^{q}\bigr), \tag{1}$$

in polar coordinates $\tilde{q} = \mu^q e^{i(\psi^q + \theta_t)}$, $\tilde{k} = \mu^k e^{i(\psi^k + \theta_s)}$. Every logit is a sum of products $\mu \cos \varphi$.

A positional mechanism is a choice about one of those two factors. RoPE, MapFormer, Selective RoPE, CARoPE and Mamba-3's rotation set the phase φ. PoPE, ALiBi, the Forgetting Transformer and every gated linear RNN set the magnitude μ. xPos, MapPoPE and Mamba-3 set both. That is the whole design space of (1), and §3 explains why it is not an artefact of the parameterisation.

3 The classification, and why there are two slots

Let $\tilde{q}_t = F(t) q_t$ and $\tilde{k}_s = G(s) k_s$, so the logit is $q_t^\top F(t)^\top G(s) k_s$, and impose three conditions:

- **Linearity.** F and G act linearly on q and k.
- **Translation invariance.** $F(t)^\top G(s)$ depends only on $s - t$.
- **Continuity** in the interval.

Normalise so $F(t)^\top G(t) = I$, absorbing the constant into the keys. Then $A(d) := F(0)^\top G(d)$ satisfies $A(0) = I$ and $A(d)A(e) = A(d+e)$: the interval operators form a *one-parameter matrix group*, so $A(d) = \exp(dM)$ for a fixed generator M, and the classification of positional encodings is the classification of M up to conjugacy. This argument is due to Puranik (2026); GRAPE (Zhang et al., ICLR 2026) develops the same framework independently and uses it to prove exact special cases. The two-slot decomposition itself has now been stated at least three times independently: as multiplicative rotations plus additive logit biases (GRAPE, December 2025), as "a multiplicative transformation and an additive bias" indexed by the relative position (Vetcha, January 2026), and as the real and imaginary parts of one generator (Puranik, April 2026). It should be treated as settled rather than as anyone's contribution.

Putting M in Jordan form gives three cases and no others.

- **Diagonalisable, real eigenvalue λ.** The plane contributes $e^{\lambda d}$: exponential decay for $\lambda < 0$, NoPE at $\lambda = 0$, and a blow-up for $\lambda > 0$ that no causal design wants.
- **Diagonalisable, conjugate pair.** The real canonical form is $e^{\lambda d}\,\mathrm{rot}(\omega d)$ — *damped RoPE*. This is the Re / Im split of (1), and **there are exactly two slots because $\mathbb{C}$ has exactly two**. RetNet, xPos and Mamba-3 all live here.
- **Defective.** Repeated eigenvalues with a nilpotent part give polynomial factors d^r times the above. ALiBi is realisable this way after augmenting q and k; Jordan-RoPE (2026) is the deliberate occupant, coupling a complex eigenvalue to a nilpotent in the same block to obtain modes $d^r e^{i\omega d}$.

A remark of Puranik's is worth isolating because it is the bridge to everything content-dependent: a gate with a data-dependent decay should be read as *learning how far to advance time*, not as changing a rate. That is exactly the substitution $t \mapsto S_t = \sum_{u \le t} G(x_u)$ of §7.

4 The classical relative methods

Shaw-style relative position representations, Transformer-XL and T5's relative bias predate the rotary line and are still widely deployed. They are one of the four escape routes of §9, and the one whose route out is least obvious.

Shaw et al. add a learned vector indexed by the *clipped* lag,

$$s_{ij} = \frac{q_i^\top \left(k_j + a^K_{r_{ij}}\right)}{\sqrt{d_{\text{head}}}}, \qquad r_{ij} = \operatorname{clip}(j - i,\, -K,\, K),$$

and T5 compresses the same idea to a per-head scalar, $s_{ij} \mathrel{+}= b_{h,\,\text{bucket}(j-i)}$. Transformer-XL carries the relative form across cached segments.

They are linear in q and k and translation-invariant, and they fail only on continuity — clip and bucket are step functions, so no generator exists and beyond the table's range the operator is constant rather than continuing.

The trade-off is exact and worth stating, because it is the one design choice the rotary line silently makes. **A table can represent an arbitrary function of the lag inside its window and cannot extrapolate outside it at all; a one-parameter group is restricted to the three families of §3 and extrapolates by construction.** RoPE gives up expressiveness over short lags to buy a functional form that is defined at every lag. Whether that purchase is honoured is a separate question, and the long-context literature's own conclusion is that it frequently is not: the ability to *compute* positional features beyond the training length does not imply reliable generalisation there, which is why context extension is evaluated by short-context retention, position-wise perplexity, retrieval and long-context reasoning rather than by whether the encoding is defined.

5 The phase axis: a family of scans

Every content-dependent phase mechanism in the arena maintains a state by cumulative summation and reads it out linearly:

$$s_t \;=\; s_{t-1} + \varphi\big(x_{t-K+1}, \ldots, x_t\big) \in \mathbb{R}^m, \qquad \theta_t \;=\; A\, s_t \in \mathbb{R}^{Hn_b}, \tag{2}$$

and they differ in three numbers: the **state rank** m, the form of the increment φ, and its **temporal support** K.

Table 1: The generator family. Every content-dependent phase mechanism maintains a state by cumulative summation and reads it out linearly; they differ in three numbers.

mechanism	m	φ	K	readout A
RoPE	1	$\equiv 1$ (constant)	1	ω
MapFormer	r	$W_{\text{out}} W_{\text{in}} x$	1	$\operatorname{diag}(\omega)$
CARoPE	H	$1/(\operatorname{softplus}(xW) + 1) \in (0, 1)$	1	geometric
Selective RoPE	Hn_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	4	τI

RoPE is the $\varphi \equiv 1$ special case: $\theta_t = t\,\omega$ is a clock that advances by the same amount on every token. Everything else lets the content set the rate.

5.1 Nesting

RoPE $\subset$ MapFormer $\subset$ Selective RoPE, each inclusion *removing a constraint* rather than adding a mechanism.

MapFormer inside Selective RoPE. Four settings suffice: (i) the convolution kernel becomes a delta at the last tap, so K collapses to 1; (ii) the gate bias is set to a constant, making $\sigma(\cdot)$ a scalar g; (iii) the angle projection is set to $A/(\tau g)$ where $A = \mathrm{diag}(\omega)W_{\mathrm{out}}W_{\mathrm{in}}$; (iv) the projection bias is zeroed. The two models then compute the same θ to 1.5×10^{-5} in float32, and the constructed W_ω has rank 2 of 64.

RoPE inside Selective RoPE is unconditional: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t{+}1)b$, an exact index clock (constant to 7×10^{-9}).

RoPE inside MapFormer costs something, because φ has no bias term: a constant increment requires every embedding row to share its projection onto W_{in}. Taking $W_{\mathrm{in}} = [I_r\ 0]$ and pinning the first r embedding coordinates achieves it exactly, at a price of r of the d embedding dimensions.

Consequently **every comparison inside this column asks whether a constraint is worth keeping, and none of them is about expressivity.**

5.2 The factorisation identity

Cumulative summation is linear, so it commutes with the readout:

$$\theta_t \;=\; \omega \odot \textstyle\sum_{u\le t} W_{\mathrm{out}}W_{\mathrm{in}}x_u \;=\; \mathrm{diag}(\omega)\, W_{\mathrm{out}} \textstyle\sum_{u\le t} W_{\mathrm{in}}x_u. \tag{3}$$

All $Hn_b = 64$ of MapFormer's phase channels are linear readouts of a single r-dimensional accumulator (checked against the implementation at 1.5×10^{-5} in float32). The multi-scale frequency ladder is a *view* of one quantity, not 64 independent clocks — which is what makes the phase a coherent position code rather than a bag of unrelated oscillators.

5.3 Coherence rank

Define the **coherence rank** ρ as the dimension of the affine span of $\{\theta_t\}$, i.e. $\mathrm{rank}[\,A{\cdot}B \mid A\varphi(0)\,]$ when $\varphi(x) = Bx$. The affine part matters: RoPE's constant φ has $B = 0$, so the linear rank alone would read 0. Then $\rho = 1$ for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE.

Two thresholds bracket the useful range. Below, a packing argument: if N^D positions are carried by a ρ-dimensional code, the minimum separation of distinct positions falls at worst like

$$\min_{p\ne p'} \|\theta_p - \theta_{p'}\| \;\lesssim\; N^{-\max(D/\rho,\,1)}, \tag{4}$$

saturating at the lattice's own spacing once $\rho \ge D$. This is an upper bound and is not tight: measured exponents track it at $D \le 3$ and run steeper at $D = 5$ (-2.99 against a predicted -2.50). Above, once $\rho \ge D$ the bound says nothing further, and what remains is a question about conditioning rather than capacity — see §13.

5.4 What $K > 1$ costs

With $K = 1$ and φ linear, (2) is exactly the action of the free *commutative* monoid on tokens: the phase depends on the multiset of tokens traversed and nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a)$ and $\prod_{u\le t} R(\omega\Delta_u) = R(\omega \sum_{u\le t} \Delta_u)$. That collapse is what buys the parallel prefix scan.

A convolution with $K > 1$ makes a token's increment depend on its neighbours, so the phase stops being a function of the multiset and the map from action sequences to $SO(2)^{n_b}$ is no longer a homomorphism. Commutativity is the central computational property here, and it is also the stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable by a fixed per-token increment.

6 The magnitude axis

The other factor of (1). Three occupants, differing in whether content enters instantaneously or by accumulation.

PoPE (Gopalakrishnan et al., 2025) is defined by *deleting* a term. RoPE's real-valued q carries an implicit intrinsic phase ψ^q, and (1) shows it entering the cosine alongside the positional angle — what PoPE calls an entanglement of "what" with "where". PoPE removes the interaction $\psi^k - \psi^q$ and keeps only a content-dependent *magnitude*, $\mu = \mathrm{softplus}(\cdot) \geq 0$, with the phase a pure index clock plus a learned per-(head, frequency) constant δ_c. It is the exact complement of MapFormer, which amplifies the same phase term into a learned accumulator.

Forget gates. FoX computes a scalar $f_t = \sigma(w_f^\top x_t + b_f)$ per head and adds $\sum_{j<\ell\leq t} \log f_\ell$ to the logit. Gated linear RNNs (GLA, Mamba, Mamba-2) do the same thing inside a recurrence, $S_t = \mathrm{Diag}(\alpha_t) S_{t-1} + k_t v_t$. GRAPE proves FoX is an exact instance of its additive family. All are accumulated content-dependent magnitude, with $\log \gamma < 0$ by construction.

ALiBi is the index-driven, content-free member: a linear-in-lag penalty, realisable by a defective generator.

7 One equation

The two axes are disjoint coordinates of a single object. Put each token's contribution in *log*-polar form. Let $c^q, c^k, G : \mathbb{R}^d \to \mathbb{C}^{n_b}$ and set

$$\boxed{\tilde{q}_t = \exp\bigl(c^q(x_t) - \overline{S_t}\bigr), \qquad \tilde{k}_s = \exp\bigl(c^k(x_s) + S_s\bigr), \qquad S_t = \textstyle\sum_{u\leq t} G(x_u),} \tag{5}$$

componentwise, with $a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \mathrm{Re}[\overline{\tilde{q}_t^{(c)}}\, \tilde{k}_s^{(c)}]$. Writing $c = \gamma + i\pi$ and $G = \delta + i\eta$ with all four real,

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{e^{\gamma^q_{t,c} + \gamma^k_{s,c} + \sum_{t<u\leq s} \delta_{u,c}}}_{\text{magnitude}} \cos\Bigl(\underbrace{\pi^k_{s,c} - \pi^q_{t,c} + \textstyle\sum_{t<u\leq s} \eta_{u,c}}_{\text{phase}}\Bigr). \tag{6}$$

c is the instantaneous complex log and G the accumulated one; real parts are magnitude, imaginary parts are phase. Every mechanism in §§5–6 is a choice of which of the four real fields $\gamma, \pi, \delta, \eta$ is non-trivial.

The conjugate is load-bearing, and the two accumulators carry opposite signs. The Hermitian form $\overline{\tilde{q}}\,\tilde{k}$ *subtracts* phases and *adds* log-magnitudes. So $\mathrm{Im}\, G$ must be carried with the *same* sign on q and k to yield $\sum_{t<u\leq s} \eta_u$, while $\mathrm{Re}\, G$ must be carried *oppositely* to yield $\sum_{t<u\leq s} \delta_u$; carried with the same sign it would give $\sum_{u\leq t} + \sum_{u\leq s}$, a joint scale carrying no information about

separation. Perturbing the prefix $u \le t$ and asking whether the score moves — a relative score must not:

Table 2: Only one of the four sign combinations is interval-relative. The prefix $u \le t$ is perturbed and the score must not move.

$\mathrm{Re}\, G$	$\mathrm{Im}\, G$	relative prefix leak
opposed	**aligned**	$\mathbf{3.4 \times 10^{-16}}$
opposed	opposed	1.4
aligned	aligned	3.1
aligned	opposed	7.1

Exactly one of the four combinations is interval-relative, and it is the one (5) writes. With $\delta = \log \gamma < 0$ that slot *is* the forget gate; with η content-dependent the phase slot is MapFormer and Selective RoPE. **MapPoPE is not a third mechanism** — PoPE constrains c, the phase family constrains $\mathrm{Im}\, G$, the coordinates are disjoint, so imposing both is an immediate consequence rather than a design.

Interval-invariance is the right notion of "relative". With content-dependent increments the logit is not a function of $s - t$ and the score matrix is not Toeplitz. The property that survives, and the one that matters, is that the score depends on the tokens in $(t, s]$ and not on the prefix.

8 A map or a clock: what cancellation chooses

It is tempting to read a signed increment as simply better. It is not, and the reason decides which mechanism a task wants.

Attention sees only the interval sum $\theta_s - \theta_t = \omega \sum_{t<u\le s} \Delta_u$. What that quantity *is* depends entirely on whether the increments can cancel:

Table 3: Cancellation chooses what the interval sum measures. The two readings are mutually exclusive by construction.

	Δ signed	Δ monotone
the interval sum is	**net displacement**	**monotone in path length**
two routes $A \to B$	agree	disagree
injective in	**space**, blind to time	**time**, blind to space
what collides	same place, different times	nothing — every instant is distinct
the accumulator is	a **map**	a **clock**

These are mutually exclusive by construction, and each is *correct* for one job. A cognitive map **needs** same-place-different-time to collide — that collision is exactly what revisit prediction asks for. Language needs the opposite: a code that made token 5 and token 15 interchangeable because some "backward" token intervened would be broken, not economical.

So the monotone increment of CARoPE, GRAPE-AP and CoPE is not an oversight that the state-tracking literature caught them in. **On language a monotone clock is the right object**, which is why RoPE — $\Delta \equiv 1$, maximally monotone — is the sensible default there and why nothing went visibly wrong. The limitation appears only where the task demands a quantity that is invariant to route, which on text is essentially parity and formal-language state tracking, and on navigation is everything.

The growth exponent is a readout of which question, not a score. $\alpha \approx 1$ means the accumulator is measuring elapsed time, which is unbounded, so the code must eventually leave any range it was trained on. $\alpha \approx 0.5$ means it is measuring a random-walk position. The ideal for a bounded map would be $\alpha = 0$ — position on a torus is bounded — and nothing here achieves it, because every mechanism accumulates without wrapping. That is a design gap, not a property of the frame.

What MapFormer gave up to get the map. A signed Δ makes two visits to the same cell positionally identical. That is the point, and it also means the phase cannot represent *how long ago*. Anything recency-dependent has to get that information from somewhere else.

> **Which is a candidate account of the forget gate, whose mechanism is otherwise unaccounted for.** (The measurements cited here are set out with their protocol in §13; "unmeasured" there has the specific sense of a contrast inside its minimum detectable effect.) It adds a second, content-dependent, *monotone* accumulator in the magnitude slot — measured at $\alpha = +0.956$, ballistic, beside the phase accumulator's 0.6 — so it restores the clock that the signed phase gave up. Every otherwise-puzzling detail follows: it needs a *live* λ (frozen at zero it lands on the baseline, -0.016, unmeasured) but not *decay*; 5 of 8 seeds learn $\lambda < 0$, and those gain most ($r(\lambda, \text{gain}) = -0.516$). A negative λ still gives a monotone accumulator, counting up instead of down. On this reading it was never a forget gate — it is a clock, and the sign of λ only sets which way it runs.
>
> **Predicted, and untested:** any monotone content-dependent increment of either sign should reproduce the gain; a content-*independent* constant increment (a pure index clock) should reproduce much of it; and the gain should vanish on a task with no recency structure.

Read this way the frame's two slots are not two knobs but **two answers**: a signed phase is a map, a monotone accumulated magnitude is a clock, and a system that needs both must spend both slots. Mamba-3 claiming both deliberately is then the natural endpoint rather than a curiosity. In softmax attention the combination is so far only partial: MapPoPE pairs a content-dependent phase with a content-dependent magnitude that is *instantaneous*, so it supplies no clock at all. A content-dependent phase alongside an *accumulated* content-dependent magnitude, in softmax attention, is the configuration nothing in the table occupies.

9 The relational map

Eight axes separate everything in the arena. They are not independent, but no two mechanisms agree on all eight.

- **A1 Factor** — magnitude (Re), phase (Im), both, or neither.
- **A2 Locus** — instantaneous (c) or accumulated (G).
- **A3 Driver** — the index, the token content, a known position, or a query–key *pair*.
- **A4 Algebra** — semisimple diagonal ($\mathbb{C}^{\times}$), unipotent, non-semisimple, or non-abelian.
- **A5 Sign** — may the increment be negative?
- **A6 Rank** of the content-to-increment map.

A7 Host — softmax attention, or linear attention / SSM.

A8 Composition — parallel prefix sum, path-ordered product, or neither.

Every *abelian* row above composes by parallel prefix sum — the eighth axis, **A8**, which the table does not have a column for because all but one row share a value. There are two exceptions. LieRE's generators need not commute, so it obeys no prefix-sum law and reads its rotation directly off a position rather than accumulating; and Algebraic PE is abelian for sequences and grids but not for k-ary trees.

9.1 What leaves the frame, and how

Non-abelian transport: PaTH, DeltaNet, RWKV-7. The interval operator becomes a cumulative *product* of Householder-like reflections $I - \beta_t w_t w_t^\top$ rather than a sum of increments. PaTH proves this takes attention beyond TC^0 (on the assumption $TC^0 \neq NC^1$); DeltaNet contributes the chunkwise-parallel algorithm that makes the product trainable at scale; RWKV-7 generalises the rule to vector-valued gating and shows it recognises *all* regular languages with a constant number of layers. The price is that no $A(d)$ and no S_t exist.

Pair-dependence: CoPE. Position stops being a property of a token. A gate is computed for every query–key *pair*,

$$g_{ij} = \sigma(q_i^\top k_j), \qquad p_{ij} = \sum_{k=j}^{i} g_{ik},$$

so the distance from i to j counts only the tokens the query elects to count. The paper's own example states the idea most clearly: to measure position in *sentences*, the gate need only fire on sentence separators. Position becomes query-dependent and denominated in task-relevant units, which is exactly why no per-token θ_t exists and why CoPE cannot be written in (2). It solves selective copying, counting and Flip-Flop, on which standard encodings fail, and improves perplexity on language and code. It is also monotone ($\sigma > 0$), so §8 applies: CoPE builds a better *clock*, one whose tick is chosen by content, and not a map.

DAPE leaves by the same door and one more. It computes an additive bias as $\mathrm{MLP}(a_{ij}, b_{ij})$ from the attention score and a fixed positional prior, so it too is a function of the *pair*, and the MLP costs linearity as well. It is worth naming because of when it appeared: DAPE and CoPE are both 2024, and they made position content-dependent on the **additive** side more than a year before CARoPE, Selective RoPE, MapFormer and Mamba-3 did the same on the multiplicative side.

Broken path structure: MesaNet, Titans, TAPE. These give up increment-accumulation itself. MesaNet solves a local least-squares problem to optimality at every timestep by conjugate gradient — a prefix-global inverse rather than a running sum — and reports lower perplexity than Mamba-2 and other modern RNNs at matched scale while remaining behind transformers on in-context recall, which is the primal/dual trade-off of §11 showing up as a benchmark result. Titans replaces the linear state with a deep memory module trained at test time by gradient descent with momentum and weight decay, and scales past 2M-token needle-in-a-haystack. TAPE keeps a positional *tensor* and updates it across layers through attention and MLP blocks under permutation- and orthogonal-equivariance constraints, improving long-sequence perplexity and length-generalised addition. In none of the three is the operator applied to a stored trace a function of the interval, so the classification does not reach them.

Table 4: **The relational map.** Mechanisms that fit (5) or its unipotent extension. *Jordan-RoPE's magnitude factor is a *polynomial* in the lag, $d^r e^{i\omega d}$: its generator couples a complex eigenvalue to a nilpotent in one defective block. ‡FoX's bias $\sum \log f_\ell$ and GLA's weight $\prod \gamma$ are the same object; the unipotent label follows GRAPE's realisation, which augments q, k to embed the bias in GL, and read as $\mathbb{C}^\times$ it sits with Mamba/GLA. §Scoped: LieRE reads its rotation directly off a known position and its generators need not commute; Algebraic PE is abelian for sequences and grids but not for k-ary trees; GRAPE's additive family is "rank-1 (or low-rank)" in its own abstract, and "full" applies only to its general edge potential. ¶MapEM evaluates the same equation on the *tensor product* of the content and position spaces rather than on $\mathbb{C}^{n_b}$; see §10.

mechanism	A1 factor	A2 locus	A3 driver	A4 algebra	A5 sign	A6 rank	A7 host
NoPE	—	—	—	—	—	—	softmax
RoPE	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
ALiBi	magnitude	acc.	index	unipotent	fixed < 0	n/a	softmax
xPos	both	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
PoPE	magnitude	**inst.**	content	—	> 0	full	softmax
MapFormer	phase	acc.	content	$\mathbb{C}^\times$	**signed**	**rank** r	softmax
MapEM¶	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapPoPE	both	inst.+acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
Selective RoPE	phase	acc.	content	$\mathbb{C}^\times$	signed	full (or 8–16)	both
CARoPE	phase	acc.	content	$\mathbb{C}^\times$	$\in (0, 1)$	**1/head**	softmax
GRAPE-M	phase	acc.	**index**	$\mathbb{C}^\times$	n/a	n/a	softmax
GRAPE-A / AP	magnitude	acc.	content	**unipotent**	≥ 0	1–full§	softmax
FoX	magnitude	acc.	content	unipotent‡	< 0	scalar/head	softmax
Jordan-RoPE	both*	acc.	index	**non-ss.**	n/a	n/a	softmax
LieRE	phase	direct§	**position**	$SO(d)$, non-ab.	n/a	**tile** $b \times b$	softmax
Algebraic PE	phase	acc.	**structure**	$\mathbb{C}^\times$ §	n/a	n/a	softmax
Mamba / Mamba-2, GLA	magnitude	acc.	content	$\mathbb{C}^\times$	< 0	full/scalar	SSM
Mamba-3	**both**	acc.	**content**	$\mathbb{C}^\times$	**signed**	full	SSM

Table 5: Mechanisms that leave the frame, and which of the three assumptions each gives up.

mechanism	assumption broken	consequence
PaTH, DeltaNet, RWKV-7	the *fixed basis* (A4 non-abelian)	path-ordered product of $I+$rank-1; no S_t
CoPE, DAPE	*per-token factorisation* (A3 = pair)	$p_{ij} = \sum_{k=j}^{i} \sigma(q_i \cdot k_k)$; no θ_t
MesaNet, Titans, TAPE	the *path structure* (A8)	prefix-global inverse / deep memory / layerwise update
Shaw, T5, Transformer-XL	*continuity* (Puranik's third)	a lookup table over clipped or bucketed lag; no generator M
TEM-t	*linearity* and *commutativity*	$e_t = \sigma(e_{t-1} W_{a_t})$; see §10

Broken linearity and commutativity: TEM-t. Treated in §10, where the two independent breaks are separated.

9.2 What the mechanisms report

The map says where each mechanism sits; it does not say how well any of them works. Their own headline results, for orientation. These are not commensurable — different scales, data and benchmarks — and are given to indicate what each line takes as evidence, not to rank them.

Table 6: Reported headline results, as stated by each source.

mechanism	as reported
RoPE	the default in most open large language models; the baseline every row below is measured against
CoPE	see §9; its reported results are the toy-task set that motivates it
FoX	outperforms the Transformer on long-context language modelling, length extrapolation and short-context downstream tasks
Selective RoPE	at 370M, **NoPE attains the lowest average perplexity of the five arms** (20.78 against RoPE's 21.42); the content-dependent phase passes RoPE on downstream accuracy only after two further components (47.1 against 46.5, from a 45.8 baseline). Solves parity with a single-layer Transformer
HGRN	its own Table 11: a *data-dependent* phase is worse than a constant one; HGRN2 attributes the complex-recurrence gain to state expansion
Mamba-3	at 1.5B, +0.6 points average downstream accuracy over Gated DeltaNet, +1.8 with the MIMO variant; matches Mamba-2's perplexity at half the state size
LieRE	up to 15.1% relative accuracy over absolute embeddings on video, $\geq 1.5\%$ over RoPE-style, at 0.68% of a ViT-B's parameters
MesaNet, Titans	see §9
MapFormer	1.00/1.00/0.99 on 2D grid navigation (IID / OOD-d / OOD-s) against Mamba's 0.38/0.66/0.30; in 5D MapWM collapses to 0.75/0.50/0.35, *below* CoPE's 0.94/0.80/0.69

Two of these deserve emphasis because they cut against the direction of the field. **NoPE is not a straw man**: in Selective RoPE's own ablation it has the best perplexity of every arm tested, and Kazemnejad et al. showed a causal decoder learns an implicit position without any encoding at all.

And **HGRN's ablation is a direct negative on content-dependent phase** for language, from inside the recurrent literature. Both belong beside the positive results, and both are consistent with §8: on text a clock is all that is wanted, and an index clock is already one.

9.3 Reading the map

(i) The frame's own cells are full. Crossing A1 (magnitude / phase) with A3 (index / content) gives four cells and all four are occupied: RoPE, ALiBi, Mamba-3's rotation, FoX. GRAPE holds the index-multiplicative and content-additive cells; Mamba-3, Selective RoPE and CARoPE hold content-phase. The remaining structure is **content-dependent, signed, orthogonal and still abelian** — contextual multiplicative — which is where MapFormer, Selective RoPE and Mamba-3 sit. CARoPE occupies the same cell on every axis but the sign, where its increment is constrained to $(0, 1)$; §8 is about what that costs.

(ii) A5 is settled next door and unapplied here. Sarrof et al. (2024) observed that SSM gates are "always nonnegative ... due to exponential or sigmoid parameterizations", with a theorem: such models cannot recognise PARITY. Grazzi et al. (ICLR 2025) proved the eigenvalue version and fixed it by extending the range to $[-1, 1]$; PaTH adopts it deliberately; RWKV-7 restricts it only for training stability; and Selective RoPE's §4.2 already demonstrates single-layer Transformer parity from rotations that "model *flips*". Yet CARoPE, GRAPE-AP and CoPE are all monotone — by softplus, by construction, by sigmoid — and none cites any of that literature. §13.1 measures what the constraint costs where it can be seen.

(iii) A6 is thinner than a blank column suggests. Two rows constrain something rank-like for reasons of their own: PaTH's Householder factors are I+rank-1 by definition, and CARoPE emits one scalar per head, putting its content-to-angle map *below* MapFormer's r rather than above it. The nearest deliberate study is LieRE's generator tile-size sweep, which varies density on the *output* side and peaks in the interior at 8×8. What no one sweeps is the *input-side* bottleneck: how many dimensions of content may steer the angle.

(iv) A7 is nearly free. Selective RoPE runs in both hosts and Mamba-3 crosses over by state-space duality, so "softmax versus SSM" separates implementations rather than mechanisms.

10 Where MapEM and TEM-t sit

Two mechanisms from the cognitive-map literature are placeable, and the placements are opposite.

MapEM lifts the frame rather than leaving it. MapEM computes two attention matrices and multiplies them elementwise, $A = \mathrm{softmax}(A_X \odot A_P)$. With $(A_X)_{ts} = q^x_t \cdot k^x_s$ and $(A_P)_{ts} = q^p_t \cdot k^p_s$,

$$(A_X \odot A_P)_{ts} \;=\; \big(q^x_t \otimes q^p_t\big) \cdot \big(k^x_s \otimes k^p_s\big),$$

so the Hadamard product *is* a single bilinear form — on the tensor product of the content and position spaces, of dimension $d_x d_p$ rather than d (verified to 1.9×10^{-6} in float32). Everything in §7 applies with c and G valued in $\mathbb{C}^{d_x d_p}$.

That is sharper than "an AND-gate rather than an OR-gate", and it says what the lift costs: a rank-one tensor $q^x \otimes q^p$ can express content-*and*-position but not content-*or*-position, and there is no fallback channel when either factor is uninformative. It predicts the observed sensitivity to how the position channel is initialised — a single shared p_0 beats separate q^p_0, k^p_0 by $+0.358$ on a matching task (3/3 seeds) — because with no additive fallback a mis-set position factor multiplies the whole score instead of being outvoted.

TEM-t is the same machine with two conditions removed. TEM-t (Whittington et al., 2022) generates position recurrently, $e_t = \sigma(e_{t-1} W_{a_t})$ with σ a ReLU and W_a a general per-action matrix, then sets $Q = K = W_e e$ and $V = W_x x$; content is absent from the logit entirely and enters only through the values.

Its logit is $e_t^\top W_e^\top W_e\, e_s$. Writing $e_s = W_{t\to s} e_t$ and $e_t = W_{0\to t} e_0$, with W_e orthogonal this is $e_0^\top W_{0\to t}^\top W_{t\to s} W_{0\to t} e_0$ — **the interval operator conjugated by the prefix**. Conjugation is trivial exactly when the group is abelian:

Table 7: TEM-t's logit is interval-relative only when the transition group is abelian and there is no nonlinearity — which is MapFormer's configuration.

transition group	logit depends on the interval alone?
abelian (block rotations), no σ	**yes**
general $SO(d)$, no σ	no
abelian, with ReLU	no
general $SO(d)$, with ReLU	no

MapFormer is precisely the surviving row: $SO(2)^{n_b}$ is abelian by construction and there is no nonlinearity on the path. TEM-t breaks both, and either break alone suffices.

This is the mechanical account of a large empirical cost. No group law means no S_t, hence no parallel prefix scan, hence a sequential loop: at $L = 2048$ a faithful TEM is 120× slower than at $L = 128$ and 1632× slower in absolute terms than a MapFormer of 20× *more* parameters. The constant includes interpreter overhead; the shape is the missing group law. TEM-t is 2022 and MapFormer 2025, so on this axis the later contribution is the *restriction* — abelian and linear — that buys the scan and the relative law.

11 The memory reading: what the frame looks like from the other side

A large literature — fast weight programmers, linear attention, gated linear RNNs, state-space models and their hybrids — studies the same equation while asking a different question. It is worth stating the correspondence, because the two literatures partition the design space almost perfectly and neither says so.

One primitive, two forms. The outer product $v_u \otimes k_u$ binds a value to a key, and superposing bindings gives an associative memory read by a query. The object appears as Kohonen's correlation-matrix memory (1972), Schmidhuber's fast weight programmer (1992), the recurrent form of linear attention (Katharopoulos et al., 2020), the hidden state of a scalar-A state-space model, Smolensky's tensor-product role–filler binding (1990), and — via the dual form of Irie et al. (2022) — the weight matrix of any linearly-parameterised layer trained by gradient descent. Schlag et al. (2021) made the linear-attention correspondence explicit. This gives every mechanism two readings:

$$\underbrace{W_t = \textstyle\sum_{u\le t} A(t-u)\, v_u k_u^\top, \quad y_t = W_t q_t}_{\textit{primal}\text{: compress, bounded state}} \qquad \Longleftrightarrow \qquad \underbrace{a_{ts} = \tilde{q}_t \cdot \tilde{k}_s \text{ over kept items}}_{\textit{dual}\text{: keep and attend}}$$

The positional operator is the memory's transport kernel. $A(d)$ is what happens to a stored trace as it ages d steps, and §3 says there are two things it can do:

Table 8: The two slots, read as a memory. $A(d)$ is the transport kernel applied to a stored trace as it ages.

slot	positional reading	memory reading
$\mathrm{Re}\, G$	magnitude / decay	**how fast the trace is forgotten**
$\mathrm{Im}\, G$	phase / rotation	**how the trace is re-addressed as it ages**

It also identifies A8 with the primal/dual choice: **a positional mechanism admits a parallel prefix scan exactly when the memory admits a bounded recurrent state**, because both are the group law $A(d)A(e) = A(d + e)$. That is one condition, not two.

The two literatures have divided the slots without noticing. The fast-weight literature owns $\mathrm{Re}\, G$ exhaustively. Irie and Gershman's review (TMLR 2025) categorises the whole modern family by *decay type* — RetNet a constant scalar, Mamba-2 a scalar, GLA a rate per row of the fast weight — and the words "rotary", "RoPE" and "positional" do not occur in it. The positional-encoding literature owns $\mathrm{Im}\, G$ and treats decay as a side-condition. Mamba-3 is the first design to claim both deliberately, and it arrives from the memory side, stating the crossing as a proposition on "data-dependent RoPE equivalence". The hybrid quadratic-linear transformers of Irie et al. (2025) use RoPE, but only as an off-the-shelf component inside the attention half rather than as a design axis.

Positional mechanisms are key-side by construction, which makes the frame a theory of addressing. Gershman, Fiete and Irie (2025) argue that what distinguishes key–value memory from classical similarity-based retrieval is that it holds *separate* representations for storage and for retrieval, so the two can be optimised independently — fidelity for values, discriminability for keys — and that the binding constraint on memory performance is addressing rather than capacity.

Every mechanism in Table 4 acts on q and k and none of them touches v. Read through the correspondence above that is not a convention: **a positional mechanism is an addressing rule**, and $A(d)$ says how the address of a trace changes as it ages. The two slots are then two ways for an address to move: $\mathrm{Im}\, G$ transports it around a torus, so a trace becomes retrievable again at a predictable displacement; $\mathrm{Re}\, G$ makes it fade, so the trace becomes unretrievable at a rate the content chooses. Nothing in the frame can alter what was stored, only where it can be found.

Why decay is compulsory on one side of the line and optional on the other. A visible asymmetry in Table 4 is that every SSM row carries $\mathrm{Re}\, G < 0$ while most softmax rows leave the magnitude slot empty. Schlag et al. (2021) supply the reason: an additive outer-product memory has a finite *capacity*, and once the sequence length exceeds it the model enters an overcapacity regime in which stored associations interfere. In the primal form decay is therefore capacity management and not a modelling preference — something has to evict. In the dual form the items are kept separately and the softmax renormalises, so nothing saturates and a decay term (ALiBi, FoX) is an inductive bias one may decline. The same equation, and the slot is mandatory on one side and free on the other.

That also sharpens what the delta rule is for. Schlag et al. introduce it precisely because purely additive updates cannot *edit* a stale association, only bury it; the rewrite is a capacity fix before it is an expressivity fix, and the expressivity — state tracking, parity — comes along with it.

The non-abelian family is the rewritable-memory family. The first row of the out-of-frame table is not a collection of exceptions. DeltaNet, PaTH and RWKV-7 replace the *additive* Hebbian update with a *delta rule*,

$$W_t = W_{t-1} + \sigma(\beta_t)\big(v_t - W_{t-1}\phi(k_t)\big) \otimes \phi(k_t),$$

whose state-transition matrix $I - \sigma(\beta_t)\phi\phi^\top$ is a Householder-like reflection rather than a scalar. Memory becomes rewritable — stale associations are corrected instead of accumulated — and the price is exactly the group law: reflections do not commute, so no $A(d)$ and no S_t exist and the composition is a path-ordered product. **What that family trades away is the positional structure, and what it buys is a memory that can overwrite.** It is also where the sign axis of §9(ii) is deliberately exercised: with $\sigma = 2\,\text{sigmoid}$ the eigenvalues lie in $(-1, 1)$ *including negative values*, which is what unlocks parity and modular state tracking.

And MapEM's lift is a binding. The tensor product $q^x \otimes q^p$ of §10 is Smolensky's role–filler binding: content as filler, position as role. Read from the memory side, MapEM stores each observation bound to its place rather than superposing the two channels additively — which is the same trade the primal/dual axis makes, one level down.

This is a change of reading, not of mechanism. It says where to look for the unoccupied designs: the cells that are empty in one literature are usually the ones the other has already filled.

12 The readout axis

§5 treats the readout $A = \text{diag}(\omega)$ as fixed. A large and practically important literature varies exactly that, and it is disjoint from everything above: Position Interpolation, RoPE scaling laws, NTK-aware scaling, Dynamic NTK, NTK-by-parts, YaRN, LongRoPE and LongRoPE2 all rescale the frequency ladder so a model trained at one length runs at another. Every one of them is index-driven, so on Table 4 they occupy a single cell — but they vary something the eight axes do not name.

The critical dimension. The argument that organises that line is worth importing. Channel r has wavelength $T_r = 2\pi b^{2r/d_{\text{head}}}$. High-frequency channels complete many cycles inside the training length L_0 and are trained through their whole phase range; low-frequency channels may not complete even one. The boundary

$$r_{\text{crit}} \;=\; \min\{\, r : T_r \ge L_0 \,\}$$

separates them, and beyond the training length the channels past it enter phase intervals never seen in training. For LLaMA-2 ($d_{\text{head}} = 128$, $L_0 = 4096$) that is 36 of 128 dimensions. This is a mechanism for length-extrapolation failure that is about *which channels were trained*, not about capacity or precision.

13 Measurements

The axes above are separable only where position must be reconstructed from the input rather than read off the index, which is why the measurements below are on navigation and algorithmic tasks rather than on text.

Setup. The primary task is **torus revisit prediction**: an agent takes actions on an $N \times N$ torus, observations are aliased across cells (16 types over N^2 cells, half of them blank), the token stream interleaves actions and observations, and the model predicts the observation at *revisited* cells only. Unless stated otherwise: $N = 64$, one layer, two heads, $d_{\text{model}} = 128$, $n_b = 32$, trained at sequence length 128 for 300 epochs with warmup and cosine decay at $\text{lr} = 10^{-3}$, and evaluated on a held-out observation map. "Accuracy" is next-token accuracy at revisited positions, against a measured always-predict-blank floor of 0.506. Two further tasks appear where noted: **parity** at sequence length L, and **blind continuation**, in which the model explores with observations revealed and then predicts the observation at each cell with observations withheld (chance $1/16 = 0.0625$).

Standard errors are over seeds; MDE is $2.8\,\text{sd}/\sqrt{n}$, and a contrast inside its MDE is reported as *unmeasured* rather than as a null. Contrasts are paired per seed, and arms compared against one another were trained in one batch.

13.1 The sign of the increment

Six arms $\times$ 12 seeds, one batch, $r = 4$ throughout, **parameter count identical at** 204,757 — the constraint adds nothing and does not touch the rank.

Table 9: The sign ablation, **raw** accuracy. Six arms $\times$ 12 seeds, one batch, identical parameter count (204,757). The construction control `Vanilla_r4`, omitted here for width, is within +0.006 of `signed` at every length — *unmeasured*, which is the condition for the batch to be readable at all. Read this table beside the loss-matched contrasts below: raw and loss-matched point the same way for the signed-vs-monotone comparison and opposite ways for monotone-vs-index.

arm	increment	T=128	T=512	T=1024	final loss
signed	$W_{\text{out}} W_{\text{in}} x$	1.000	0.978	0.922	0.0002
$\lvert \cdot \rvert$	$\lvert W_{\text{out}} W_{\text{in}} x \rvert$	0.946	0.675	0.558	0.171
softplus	GRAPE-AP form	0.977	0.798	0.584	0.068
CARoPE	$1/(\text{softplus}(\cdot) + 1)$	0.900	0.809	0.645	0.293
RoPE	index	0.799	0.449	0.345	0.784

The raw column is not the contrast of interest, and says so plainly: the monotone arms fit far worse (0.171, 0.068, 0.293 final loss against 0.0002), so a raw comparison against an index arm that fits worse still (0.784) reports convergence rather than representation. **At matched training loss, signed path integration beats an index code by** +0.123 **and** +0.195 **at** $T = 512$ **and** $T = 1024$ **(**12/12 **seeds), while monotone path integration beats it nowhere** — −0.028 at $T = 128$ and unmeasured beyond. Note also that the index arm sits at or below the measured always-predict-blank floor of 0.506 at both OOD lengths, so its raw margins are not interpretable as skill. The entire measured value of a content-dependent phase over a plain index clock, on this task, requires the increment to be signed.

The learned code shows why. The opposition score $\|\Delta(+x) + \Delta(-x)\| / \overline{\|\Delta\|}$ is 0 when opposite actions cancel exactly and 2 when they are identical:
CARoPE's published parameterisation reaches 1.98 of a possible 2.00, with its north and east axes all but collapsed onto one another. These models do not merely score worse — they cannot represent a -1 action, and the probe confirms they do not.

At the training length the signed baseline is at 1.000 ± 0.000, so accuracy there cannot show a deficit of any size. The training-length effect is visible in the *loss* instead: every constrained arm fits strictly worse on 12/12 seeds at identical parameters.

Table 10: The learned action code. Opposition is 0 when opposite actions cancel exactly and 2 when they are identical.

arm	oppose_x	oppose_y	$\lvert\cos(N, E)\rvert$
signed	**0.125**	**0.106**	0.218
$\lvert \cdot \rvert$	1.849	1.855	0.587
softplus	1.905	1.885	0.723
CARoPE	**1.981**	**1.977**	**0.930**

This is a replication in a new regime. The constraint and its consequence for parity are established (§9(ii)); what is new is navigation rather than formal languages, and an isolation in which $|\Delta|$ against Δ is a single operation at identical parameter count.

13.2 Rank

The bottleneck r is meant to be the dimensionality of the action space — for a 2D grid, a 2-vector — which makes $r = 2$ sufficient *by construction*. It is not sufficient in practice. Against $r = 2$ at 8× the training length ($T = 1024$ against a training length of 128; 8 seeds):

Table 11: The rank sweep against $r = 2$, 8 seeds. A step at $r = 2$, flat from $r = 4$ to $r = 32$.

	params added	T=512	T=1024
$r = 4$	+384	+0.038	+**0.085** (8/8, $t\,3.57$)
$r = 8$	+1,152	+0.037	+0.091
$r = 16$	+2,688	+0.028	+0.079
$r = 32$	+5,760	+0.038	+0.095

This is a **step at $r = 2$, not a capacity slope**: flat from $r = 4$ to $r = 32$. The cause is conditioning, not capacity. Given four dimensions the model still places its actions in a 2-plane (100.0% of spectral energy; 99.96% even at $r = 32$), vindicating the dimensional argument — what fails is the *basis* that $r = 2$ is forced into:

Table 12: The learned basis at $r = 2$ and $r = 4$. The deficit is conditioning, not capacity.

	opposition ($N{+}S \approx 0$)	$\lvert\cos(N, E)\rvert$	obs/action norm
$r = 2$	0.495	0.783	0.139
$r = 4$	**0.092**	**0.175**	0.042

At $r = 2$ opposite actions fail to cancel by half the action scale and north and east are nearly parallel. The seed standard deviation also falls from 0.064 to 0.012 at $T = 1024$, which matters because it sets every detectable effect size.

Two further observations. The deficit does *not* grow with the dimensionality of the world: at $D = 5$, $r = 2$ is unimpaired (0.896, its best of three conditions), so the packing account of (4) does not explain it. And an unconstrained generator cannot be projected below rank ≈ 16 post hoc while $r = 4$ trains fine — trained-at-rank and truncated-to-rank are different objects.

13.3 What the other constraints are worth

Selective RoPE's three additions to MapFormer's generator, measured on the torus at $T = 1024$ and on a parity task at $L = 128$; increments against MapFormer at $r = 2$.

Table 13: Selective RoPE's additions to MapFormer's generator, against MapFormer at $r = 2$.

constraint removed	params	parity L=128	torus T=1024
$K = 1 \to 4$ (causal conv)	+193	−0.020	−0.064 (t −1.99)
$m = r \to Hn_b$ (full rank)	+7,873	−0.020	+0.058 (7/8)
φ linear → gated	+8,193	−0.030	+0.086 (8/8)
all three	+16,385	−0.009	+0.048 (t 1.36)
$r = 2 \to 4$, *constraint kept*	**+384**	—	**+0.085** (8/8)

Two readings. First, **the two ≈8k-parameter knobs are statistically indistinguishable from each other** (+0.018 and +0.028, both inside their MDEs), so what they buy is most plausibly capacity rather than either named mechanism — and widening the bottleneck MapFormer already has buys the same thing for 384 parameters, 21× cheaper. Second, the convolution is the only knob negative on both tasks, consistent with §5.4: $K > 1$ costs the homomorphism.

These arms replace the readout $\mathrm{diag}(\omega)W_{\mathrm{out}}$ by τW_2. For the conv and gate rows the function class is preserved exactly — it is the construction of §5.1 — so the difference is initialisation (a geometric frequency ladder against a default draw) plus one scalar; those effects are attributable up to initialisation. The two full-rank rows also add weight normalisation and a bias, so their function class genuinely differs.

A separate probe rules out the natural mechanism for the gate. If the sigmoid worked by suppressing observation tokens — which MapFormer must instead learn to send to $\Delta \approx 0$ — the gate would be near zero on observations. It is 0.416, and the action/observation ratio is 1.35× on the torus where the gate helps and 1.54× on parity where it hurts.

13.4 The navigation regime

Every paper in §9 evaluates on language, recall, state tracking, or — for LieRE and Algebraic PE — images, video and trees. None runs grid navigation, and it is the regime in which these axes are distinguishable at all.

The basic separation is large. On a task where the model explores with observations revealed and then continues blind, predicting the observation at each cell, path integration scores 0.730 ± 0.247 ($n = 5$, 64^2) against 0.154 ± 0.018 for an index code, chance 0.0625, with no seed overlap. The axis is path integration and not the encoding: PoPE with path integration scores 0.847 against 0.117 for PoPE with an index. On the standard revisit task, re-derived from the per-seed data ($n = 8$), the position main effect is +0.468 and the encoding main effect is +0.001. No ratio is quoted: with a denominator that close to zero the quotient is not a stable quantity, and published values for it range from 40× to 370× depending on which seed set and averaging convention is used. The defensible statement is that one factor moves the result by roughly half and the other does not move it measurably.

Two boundaries are worth stating. The effect is not a property of navigation as such but of *map extent*: at matched observation aliasing it is −0.010, +0.015 and +0.305 for 32, 128 and 512 occupied cells — a threshold, not a gradient. And it does not survive rotation-based action spaces: under turn/turn/forward, where displacement depends on accumulated heading, the effect falls from

+0.438 to +0.050, which is §5.4's commutativity limitation made concrete. Recording the absolute displacement instead of the commanded turn restores it to +0.488.

13.5 Anchors

Table 14: Anchors: what each ingredient costs and buys on the torus task.

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
Signed increment	sign of Δ	**0**	largest per parameter: removing it costs −0.215/−0.280 at T=512/1024 (12/12)
Bottleneck rank	state rank ρ	+384 (+0.19%)	+0.085 at T=1024 for r=2 → 4 (8/8); flat to r=32
PoPE	magnitude μ	+320	+0.027 *with* path integration; both arms on the floor without it

13.6 Where the length dependence does not come from

Transposed to a content-dependent phase the coordinate is the accumulator rather than the index, which gives a prediction: the benefit of the OOD-helping mechanisms should localise to the channels whose period exceeds the trained range of S_t, and ablating those channels should cost *less* at extrapolated length because they are already being read out of distribution.

The precondition holds — at the training length 8–17 of 64 channels never complete a cycle of their own period, and at 8× that length only 1–4 do — so the mechanism could operate. It does not. Ablating the 32 lowest-frequency channels is free at training length (−0.001) and costs 0.14–0.25 at $T = 1024$, on every arm whose baseline is off the floor. **Those channels are more load-bearing at extrapolated length, not less**, which is the opposite of the prediction. Damage is localised in frequency — high-frequency ablation costs 0.42–0.57 — but the plainer reading wins: the low-frequency channels carry position at map scale, and coarse position matters more when the walk covers more of the map.

13.7 The accumulator's growth law

All four arms start from the same accumulator range at training length (88–94) and separate entirely by how fast it grows with sequence length, range(S) $\sim T^{\alpha}$:

Table 15: The accumulator's growth law. All four arms start from the same range at training length and separate by exponent.

arm	T=128	T=1024	α	opposition score
signed, r=4	91.8	269.7	**0.518**	0.125
signed, r=4 (rank sweep)	94.1	279.7	**0.524**	0.092
signed, r=2	88.5	320.5	0.619	0.495
monotone, r=4	91.5	649.8	**0.943**	1.849

A signed, well-conditioned increment gives $\alpha \approx 0.52$: the accumulator performs a **diffusive random walk**. A monotone one gives $\alpha \approx 0.94$: **ballistic drift**. The last column is the opposition score of §13.1; across the four arms $r(\text{opposition}, \alpha) = +0.9995$.

This is one mechanism for two separate results. How completely opposite actions cancel sets whether the accumulator diffuses or drifts; that sets how fast it leaves the range it was trained on; and that sets the rate of degradation with length. A skewed rank-2 basis (opposition 0.495) and a monotone increment (1.849) are the same failure at different severities. The relation is correlational across four arms, and $\alpha \approx 0.5$ for a signed walk is unsurprising in itself; what it rests on is the spread to 0.94 and the agreement with an independently measured quantity.

14 Choosing a mechanism

What a task demands of a positional code, which mechanisms supply it, and which trade-offs are forced rather than incidental.

14.1 Three hard trade-offs

(T1) Commutativity buys the scan and costs path-dependence. With $K = 1$ and a linear increment the phase depends on the multiset of tokens traversed, the prefix product collapses to a prefix sum, and inference is $O(\log T)$ parallel. The same property makes displacement-under-accumulated-heading unrepresentable: a fixed per-token increment cannot express "forward" when forward depends on where you are facing. Measured both ways — the position effect falls from +0.438 to +0.050 under turn/turn/forward actions, and a mechanism that abandons the group law pays for it in the scan (§10). You may have one or the other.

(T2) A signed increment buys a map and gives up a clock. The accumulator's growth exponent is set by how completely opposite actions cancel, but the exponent is a symptom and the substance is §8: a signed increment makes the interval sum a *net displacement*, invariant to route and blind to elapsed time, and a monotone one makes it a *path length*, injective in time and blind to place. Neither dominates. On navigation the signed form wins outright (at matched training loss the monotone one does not beat a plain index clock anywhere); on language the monotone form is the correct object, and the signed one would be discarding the very distinction position exists to make.

(T3) Compressing the memory makes decay compulsory; rewriting it forfeits T1. Both halves are argued in §11: the primal form saturates so something must evict, while the dual form renormalises and may decline a decay term; and a memory that must *overwrite* needs a delta rule, whose transition matrix is a reflection rather than a scalar.

14.2 What a task demands, and what supplies it

14.3 What follows for choosing a model

On language, most of this is invisible, for the reason §8 gives: order is nearly all that is demanded, and a clock is the right object for it.

On navigation and any task with a metric structure, the sign and the conditioning of the increment are the whole game. They are worth more per parameter than anything else measured here, and both dominate the choice of positional *encoding* (§13).

Table 16: What a task demands of a positional code, and what supplies it.

if the task needs ...	it requires	and these supply it
order only — which token came first	any monotone clock	RoPE, ALiBi, CoPE, CARoPE, GRAPE-AP, every gated linear RNN
signed displacement — parity, net motion, state tracking	a signed increment	MapFormer, Selective RoPE, Mamba-3; DeltaNet/PaTH/RWKV-7 with $\beta > 1$. **Not** CARoPE, GRAPE-AP or CoPE
absolute location in a bounded map	signed **and** well-conditioned ($r \geq 4$)	MapFormer at $r \geq 4$; $r = 2$ degrades because its basis is skewed, not because it lacks capacity
holding position far past the training length	a diffusive accumulator, $\alpha \approx 0.5$	follows from cancellation (T2); nothing in Table 4 bounds the accumulator explicitly, so this is the only route available
path-dependent actions (turn/forward)	non-abelian transport, **or** an allocentric recoding of the input	PaTH, DeltaNet, RWKV-7, TEM-t — none evaluated on navigation. Recoding the action to absolute displacement restores +0.488 and keeps the scan
overwriting stale associations	a delta rule	DeltaNet, PaTH, RWKV-7; forfeits the parallel scan
exact recall over a window	the dual form; keep the items	softmax attention. A lossy summary is not a sufficient statistic, which is why hierarchy loses here and wins on compositional transfer
bounded state, linear time	abelian transport	every row of Table 4; measured 2.6–3.3× over a 16× length increase

One combination is ruled out rather than merely absent. A mechanism with a bounded recurrent state, signed transport *and* rewritable memory does not exist, and T1 says the last two cannot both be had while keeping the parallel scan.

15 Conclusion

Three assumptions — linearity, translation invariance, continuity — reduce the space of positional mechanisms to the classification of a single generator, and the generator has exactly two live parts. Everything in the arena is a choice about where content enters those two parts, and everything that appears to escape does so by giving up one of the three assumptions, at a price that can be named: the classical relative methods give up continuity and with it any extrapolation; the delta-rule family gives up commutativity and with it the parallel scan; CoPE and DAPE give up per-token factorisation and with it any θ_t at all; and MesaNet, Titans and TAPE give up the accumulation structure itself.

The two parts are not two knobs. A signed accumulated phase is a map and a monotone one is a clock, and a system that needs both must spend both slots. That is why the monotone increments of CARoPE, GRAPE-AP and CoPE are correct on language and fail on navigation, and why the sign of the increment — which costs nothing, and which no paper in the positional-encoding literature varies deliberately — is worth more than any other ingredient measured here.

What the frame does not yet explain is the axis on which several mechanisms independently help: extrapolated length. The accumulator's growth law accounts for two of them and demonstrably not for the others — and no mechanism in the table bounds the accumulator at all, including the one that was assumed to. That is the gap worth building into.

16 Open questions

- **The input-side rank.** $r = 2$ is the worst setting and $r = 4$ fixes it for 384 parameters, but nothing explains why the optimiser finds a skewed basis at $r = 2$ and a well-conditioned one at $r = 4$ when the learned solution occupies a 2-plane either way.
- **The OOD-length axis, now partly explained.** The accumulator's growth exponent accounts for the sign and rank results together (§13.7), and it predicts that bounding the accumulator should help at length — which is Measured, neither the forget gate nor PoPE changes α at all (both inside their MDEs), so at least one other cause is at work. §8 proposes one for the forget gate — it restores the clock a signed phase gives up — with three untested predictions. The direct test of the growth law itself is to vary α deliberately and check that OOD degradation follows.
- **Defective generators.** Jordan-RoPE occupies the cell; what polynomial factors in the lag are *for* remains open, and no navigation evaluation exists.
- **NoPE.** A causal decoder learns an implicit position without any operator at all, and in at least one controlled ablation it has the best perplexity of every encoding tested (Table 6). The frame classifies operators and has nothing to say about the case where the right operator is the identity; when an encoding helps on text, how much of the gain is over NoPE rather than over RoPE is usually not reported.
- **Sign on language.** A monotone clock cannot represent a -1 action, and on navigation that costs everything. On text HGRN's own ablation finds a data-dependent phase worse than a

constant one. Whether the sign matters there, and for what, is untested.

- **Non-commuting action spaces.** Rotation-based actions defeat a fixed per-token increment. Allocentric recoding sidesteps it wherever heading is known; the non-abelian mechanisms (PaTH, DeltaNet, RWKV-7) address it directly and have never been evaluated on navigation.

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